

Complex Engineering Problem (CEP)

Computer Programming in Python

Department of Electronic Engineering

The Islamia University of Bahawalpur, Pakistan

Instructor: Dr. M. Abbas Abbasi

Project Overview

You will work in pairs (maximum 2 students) to build a **simple web application** from start to finish. You will choose your own project idea, design a database, create a backend with Python and FastAPI, build a frontend with React, and deploy it online using free tools.

What you will build:

- A simple web app with a database (SQLite)
- A backend API using Python and FastAPI
- A frontend using React
- The final app will be deployed online (free hosting)

Example project ideas: Student attendance system, Lab equipment booking, Simple inventory tracker, Complaint management system

Course Learning Outcomes (CLOs) Mapping

Your project must demonstrate the following CLOs from the course outline:

- **CLO-1:** Create Python programs to solve real-world problems, incorporating advanced concepts like inheritance, polymorphism, exception handling and file I/O.
- **CLO-2:** Evaluate and assess emerging Python technologies and libraries and make informed decisions about their suitability for specific programs.
- **CLO-3:** Perform code development and debug tasks on Python development environment to program specific tasks.
- **CLO-4:** Complete clear and concise documentation for code and projects, including code comments and README files.
- **CLO-5:** Infer the importance of project planning and scheduling in achieving project success.

Note: Each week addresses specific CLOs - see weekly plan below.

General Rules

- Work in pairs (2 students maximum)
- You can use tutorials but must understand every line of code
- Present your progress every week (5-7 minutes)
- Keep your code on GitHub from Week 1
- Final submission includes: working deployed app + code + README + presentation

Grading (Total 100%): Working app and deployment (35%), Code quality and Python concepts (30%), Database and API design (20%), Documentation and presentation (15%)

Week-by-Week Plan

Week 1: Project Idea Presentation

What to do this week:

- Form your pair (2 students)
- Choose a project idea (something useful for your home, department or university)
- Draw a simple table showing what data you will store (e.g., Student: name, roll number, class)
- Install Python, FastAPI, Node.js, and React on your computer
- Create a GitHub account and make a new repository for your project

What to submit/present:

- 1-page document explaining: What problem you are solving, who will use it, what data you will store
- Simple drawing of your database tables
- Show that Python and React are installed on your computer

CLOs: CLO-4 (documentation), CLO-5 (project planning and scheduling)

Week 2: Database and Backend

What to do this week:

- Create your SQLite database with 1-2 tables
- Add 5-10 sample records to your database
- Write a simple FastAPI program that:
 - Connects to your database
 - Has one endpoint to show all records (e.g., GET /students)
- Test your API using FastAPI docs (go to <http://localhost:8000/docs>)
- Push your code to GitHub

What to submit/present:

- Show your database file and the tables inside
- Demonstrate your API working (show the /docs page and run the endpoint)
- Explain how your Python code connects to the database

CLOs: CLO-1 (Python programming with file I/O), CLO-3 (code development), CLO-4 (code comments)

Week 3: Frontend Display

What to do this week:

- Create a React app
- Make a simple page with a title and a table to show your data
- Connect React to your FastAPI backend:
 - Use fetch or axios to call your API
 - Display the data in a table or list
- Make sure both FastAPI and React run at the same time
- Push all code to GitHub

What to submit/present:

- Live demo: Show your React page displaying data from your database
- Explain in simple words how data flows from database → FastAPI → React → browser

CLOs: CLO-1 (Python programming), CLO-2 (evaluate technologies - React/FastAPI), CLO-3 (development environment)

Week 4: Add, Update, Delete Features

What to do this week:

- Add more endpoints to your FastAPI:
 - POST endpoint to add a new record
 - DELETE endpoint to remove a record
 - (Optional) PUT endpoint to update a record
- In React, add:
 - A form to add new data
 - A delete button next to each row
- Test everything: add, delete, and see changes
- Organize your Python code into functions
- Push to GitHub

What to submit/present:

- Demo: Add a new record from your React page and show it appears
- Demo: Delete a record and show it disappears
- Show your Python functions are well organized

CLOs: CLO-1 (exception handling in Python), CLO-3 (code development and debugging), CLO-4 (code organization)

Week 5: Extra Feature and Deployment Prep

What to do this week:

- Make your app look better: clear labels, proper layout
- Add ONE extra feature (choose one):
 - Search box to filter records
 - Show count or summary of records
 - Simple login (hardcoded username/password is fine)
- Clean up your code: remove repeated code, add comments
- Choose where to deploy:
 - Backend: Render, Railway, or PythonAnywhere (all free)
 - Frontend: Vercel, Netlify, or GitHub Pages (all free)
- Read the deployment guides for your chosen platforms

What to submit/present:

- Demo your extra feature
- Show your cleaned-up code
- Tell us which platforms you chose for deployment and why

CLOs: CLO-2 (assess emerging technologies for deployment), CLO-3 (debugging and refinement), CLO-4 (code documentation)

Week 6: Final Deployment and Presentation

What to do this week:

- Deploy your FastAPI backend to your chosen platform
- Deploy your React frontend to your chosen platform
- Make sure the deployed frontend talks to the deployed backend
- Test everything one more time
- Write a README file explaining:
 - What your project does
 - How to run it locally
 - Links to your deployed app
 - Which CLOs (CLO-1 to CLO-5) you demonstrated and how
- Prepare final presentation slides (10-12 minutes)

What to submit/present:

- Live demo using your deployed app (if internet is available)
- Explain your project idea and who would use it
- Show how React, FastAPI, and SQLite work together

- Explain what you learned and what was challenging
- Show how your project covers all five CLOs (CLO-1 through CLO-5)
- Provide: GitHub link + deployed app links

CLOs: All CLOs demonstrated: CLO-1 (complete Python application), CLO-2 (technology decisions), CLO-3 (full development cycle), CLO-4 (comprehensive documentation), CLO-5 (project completion)

Important Reminders:

- Keep it simple! The goal is to learn, not to build a complex system
- If you copy code, make sure you understand it completely
- Git commit and push your work regularly
- Test your app every week to catch problems early

Final Deliverables:

- Working deployed web application (both frontend and backend)
- GitHub repository with all code and good README
- Final presentation (slides + live demo)
- CLO mapping document (showing how you achieved CLO-1 through CLO-5)